

APPLICATIONS OF RISK SCIENCE TO AI FAIRNESS EVALUATION: PRINCIPLES, CHALLENGES, AND BEST PRACTICES

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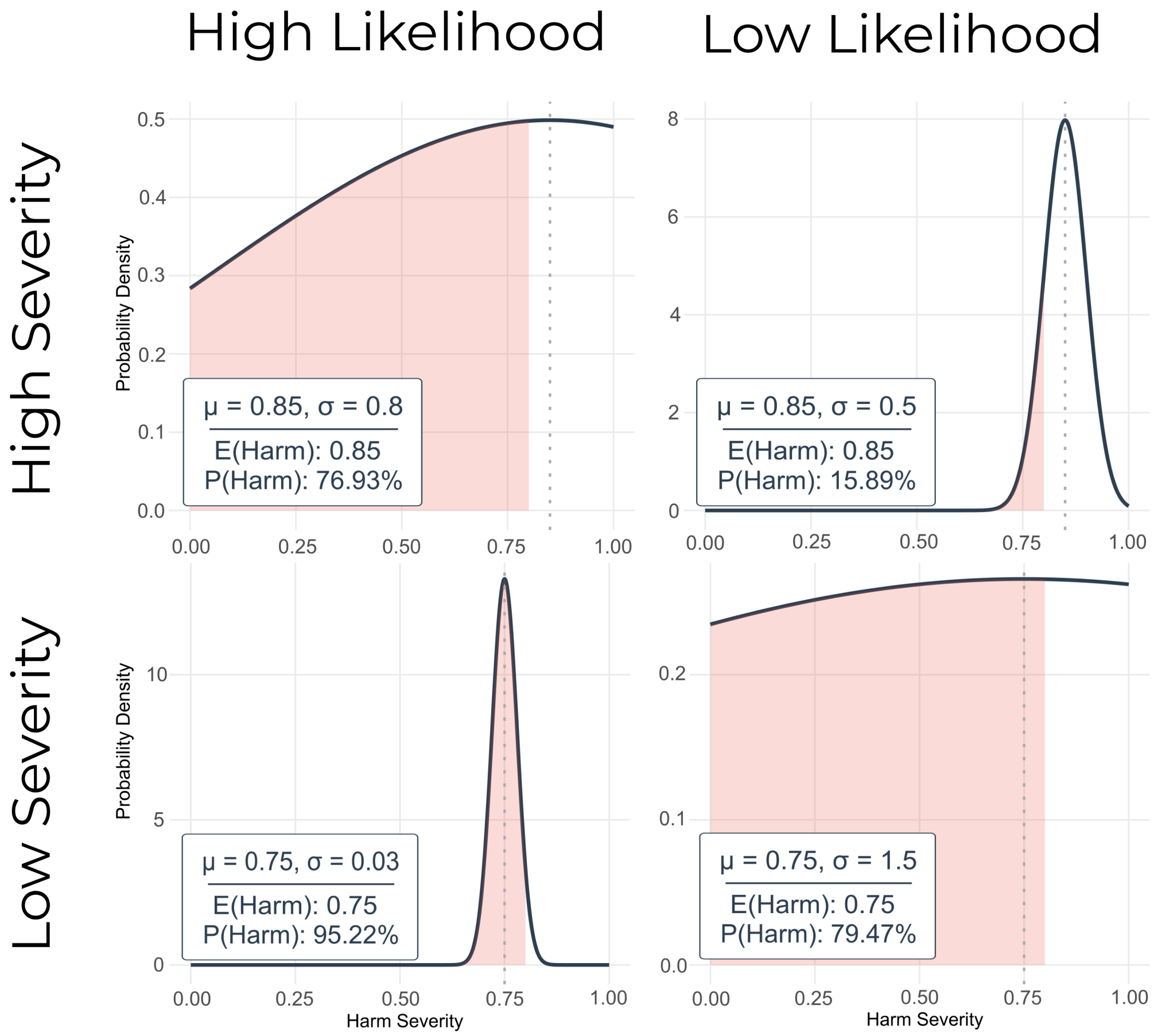


Figure 1. Illustration of how severity and uncertainty influence risk assessment. In the upper panels, severity ($\mu=0.85$) is identical, but the probability of harm H [$1(\mu<0.8)$] increases with variability σ . When severity is below the harm threshold in the bottom panels, high variability decreases the likelihood of harm.

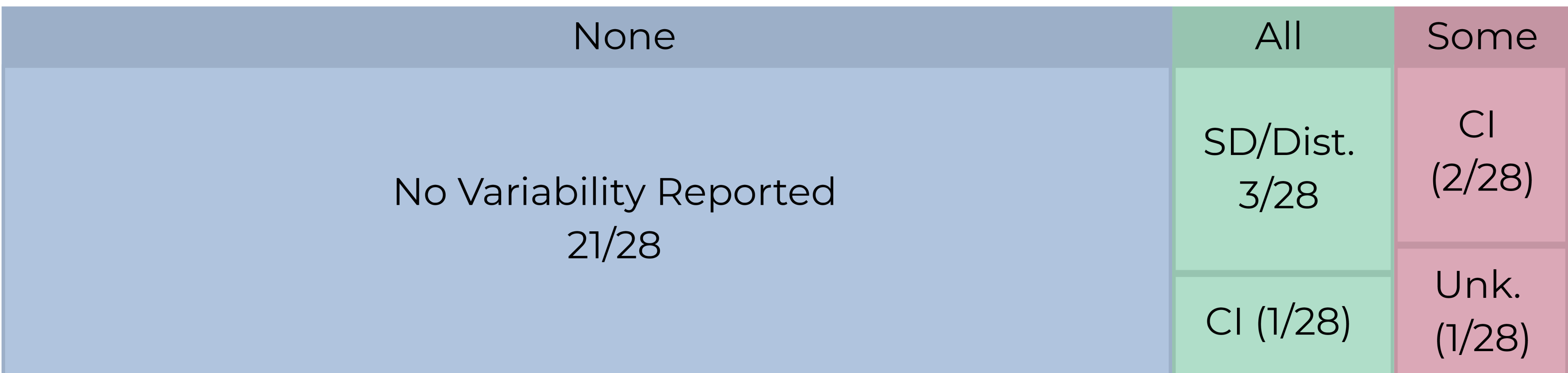


Figure 2. Among the 28 evaluations reviewed, only seven provided variability (stochastic uncertainty) estimates for their estimates of severity. Among these variability estimates there was a mix of confidence intervals (CI) and standard deviations or distributions (SD/Dist.). One study did not specify the type of variability estimate used.

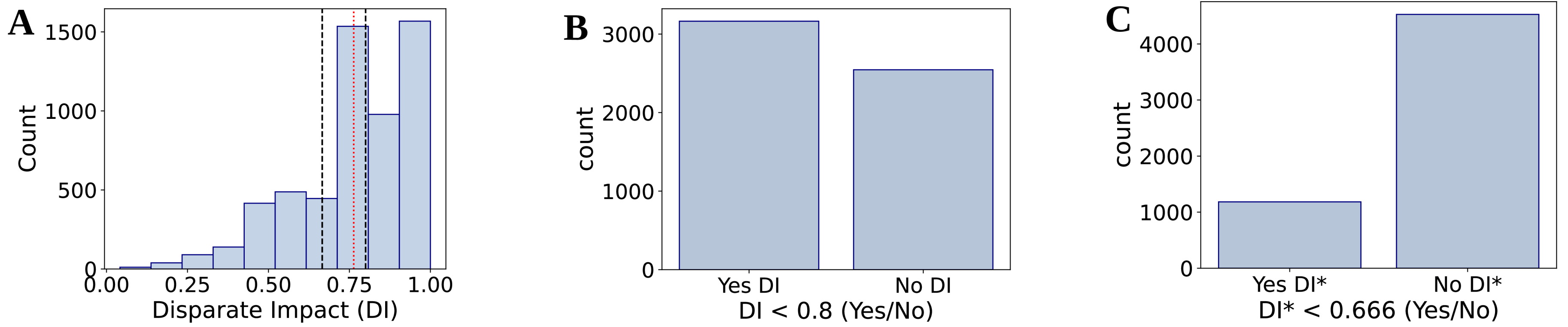


Figure 3. [A] Distribution of DI scores from [3]. The red line marks the average; the two dashed lines show possible DI thresholds at 0.666 (based on the 80% rule) and 0.800 based on statistical significance). [B and C] Resume screening scenarios that are classified as discriminatory vs. non-discriminatory using the two thresholds in [A].

PRINCIPLES

- **Risk science** provides the most updated and justified knowledge, statements, and beliefs on risk analysis [1]
- **Risk** is a function of the likelihood of a harm occurring, its **severity**, and the **uncertainty** associated with each of these
- Uncertainty can be **stochastic** or **epistemic**. The former concerns naturally occurring randomness that comes with any estimation; the latter is a result of a lack of knowledge or understanding about the phenomenon of interest.
- The overall assessment of risk depends on the likelihood and severity of harm; isolated the components provides an incomplete and/or misleading picture (Figure 1)

CHALLENGES

- We reviewed 28 evaluations of fairness in algorithmic hiring [2]
 - 75% of results had only point estimates (Figure 2)
 - Only one study of 22 reported results in terms of probability (likelihood); others used metrics of severity only
- Using [3] as a case study, we demonstrate challenges of fairness risk evaluation using disparate impact (DI) as a metric of severity
 - Stochastic uncertainty can be minimized effectively by using large sample sizes in evaluations
 - Epistemic uncertainty is large because DI has many definitions (Figure 3)
 - Additionally, making strong assumptions about real-world resume screening also increased epistemic uncertainty

REFERENCES

[1] Aven, T. 2025a. A History of Risk Science: Concepts and Principles of Risk Analysis. Taylor & Francis.
[2] Fabris, A.; Baranowska, N.; Dennis, M. J.; Graus, D.; Hacker, P.; Saldivar, J.; Zuiderveen Borgesius, F.; and Biega, A. J. 2025. Fairness and bias in algorithmic hiring: A mul- tidisciplinary survey. ACM Transactions on Intelligent Systems and Technology, 16(1): 1–54.
[3] Wilson, K.; and Caliskan, A. 2024. Gender, race, and intersectional bias in resume screening via language model retrieval. In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, volume 7(1), 1578–1590.
[4] Aven, T. 2017. Improving risk characterisations in practical situations by highlighting knowledge aspects, with applications to risk matrices. Reliability Engineering & System Safety, 167: 42–48.
[5] Flage, R.; and Aven, T. 2009. Expressing and communicating uncertainty in relation to quantitative risk analysis. Reliability: Theory & Applications, 4(2-1 (13)): 9–18.

BEST PRACTICES

- Evaluations of AI fairness risks should include:
 - Specific activities and possible consequences being evaluated relative to human values
 - Stochastic uncertainty estimates with all severity estimates
 - Epistemic uncertainties that exist in relation to the specific evaluations used
 - These can be described categorically using scales such as those in [4-5]
- We propose the **AI Risk Report Card** to present AI fairness risk evaluations (Figure 4)

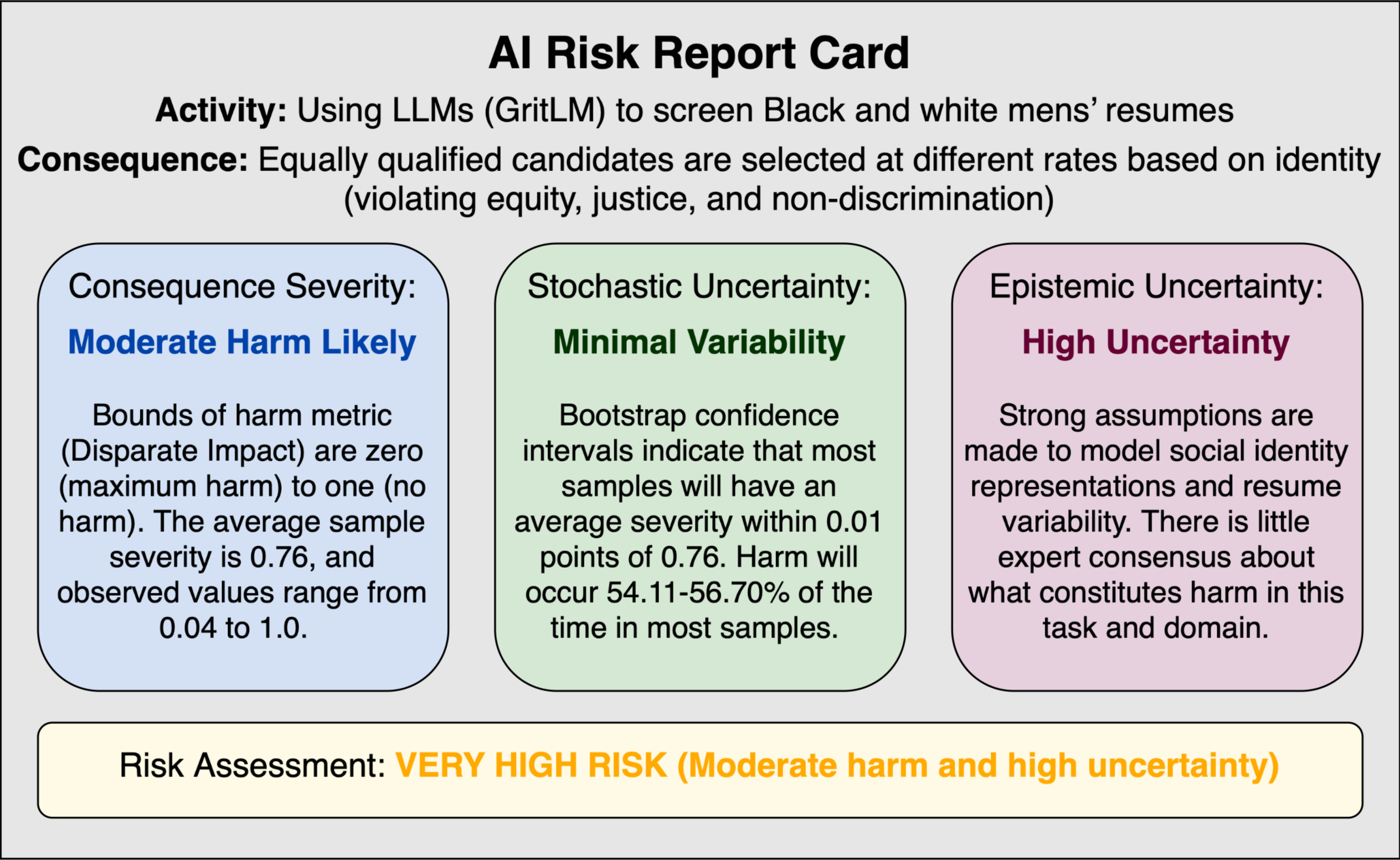


Figure 4. Example of an AI Risk Report Card for our case study results to facilitate communication between those conducting AI fairness evaluations and stdecision makers outside the evaluation community. Future work is needed to validate and refine this proposal using input from all stakeholders.

